AlmaBetter

Full Stack Data Science & AI

Curriculum

World Education Congress Award 2023

Module 1: Introduction to Computer Programming

1. Getting Started with Python	Learning Objective
	Introduction to Python Programming
	Data Types in Python
	In-Built Functions & Methods
	Indexing & Slicing
	Operators in Data Types
	Skill Mastery Challenge
2. Python Control Flow	Learning Objective
	Statements, Indentation & Conditionals
	Loops & Iterations
	Conditional & Infinite Looping
	Skill Mastery Challenge
Mid Course Assessment	**Mid Course Quiz Assessment**
	Group Learning LAB : PyCodeX
3. Object Oriented Programming	Learning Objective
	Custom Functions in Python
	Advanced Looping Concepts
	OOPs in Python
	Exception Handling
	Skill Mastery Challenge
Revision & Beyond	**Leveraging AI : Pair Programming Using ChatGPT**
	Research Paper - I
	Research paper - II
End Course Assessment	**End Course Quiz Assessment**
	Group learning LAB : LogicHack
	Capstone Project : Pythonic Challenge

Module 2: Numerical Programming in Python

1. Data wrangling Libraries	Learning Objective
	Getting Started with Pandas
	Mastering Data Wrangling
	Advanced Data Wrangling Concepts
	Data Wrangling on Different Data Formats
	Skill Mastery Challenge
2. Data Visualization Libraries	Learning Objective
	Data Visualization with Matplotlib & Seaborn
	Data Vizualization Tips & Variable Analysis
	Exploratory Data Analysis | Ecommerce Case Study
	Skill Mastery Challenge
Mid Course Assessment	**Mid Course Quiz Assessment**
	Group Learning LAB : Data Explorers
3. Data Insights with APIs & EDA	Learning Objective
	Data Modelling via APIs
	Interactive Visuals with Plotly
	Streamlit Dashboards with dynamic Interactions
	Skill Mastery Challenge
Revision & Beyond	**Leveraging AI : Data Wrangling using Pandas AI**
	Research Paper - I
	Research paper - II
End Course Assessment	**End Course Quiz Assessment**
	Group Learning LAB : AutoInsights
	Capstone Project : DataWrangle Challenge

Module 3: Relational Databases

1. Getting Started with SQL	Learning Objective
	Fundamentals of SQL Query
	Dealing With Multiple Tables
	Advanced SQL Joins
	Skill Mastery Challenge
2. In-Built Functions	Learning Objective
	Type Casting & Math Functions
	DateTime Functions
	String Functions
	Window Functions
	Skill Mastery Challenge
Mid Course Assessment	**Mid Course Quiz Assessment**
	Group Learning LAB : QueryQuest
3. Data Preparation with SQL	Learning Objective
	Complex queries using CTE & Connecting with Azure Database
	Data Exploration With SQL & Python
	Database Management & Schema Design
	Skill Mastery Challenge
Revision & Beyond	**Leveraging AI : ChatGPT for writing SQL Queries**
	Research Paper - I
	Research paper - II
End Course Assessment	**End Course Quiz Assessment**
	Group Learning LAB : SchemaStudio
	Capstone Project : The Architect Challenge

Module 4: Data Visualization Tools

1. Data Analysis using Excel	Learning Objective
	Fundamentals of Excel
	Data Exploration with In-Built Functions
	Excel Dashboarding
	PowerQuery in Excel
	Skill Mastery Challenge
2. Present like a Pro with PowerBI	Learning Objective
	Introduction to PowerBI & ETL
	Dashboarding Concepts with Power BI
	Introduction to DAX
	Advanced DAX Functions
	Skill Mastery Challenge
Mid Course Assessment	**Mid Course Quiz Assessment**
	Group Learning LAB : VizMaster
3. Present like a Pro with Tableau	Learning Objective
	Getting Started with Tableau Ecosystem
	Dashboarding & Storytelling with Tableau
	Skill Mastery Challenge
Revision & Beyond	**Leveraging AI : Creating Presentations using ChatGPT**
	Research Paper - I
	Research paper - II
End Course Assessment	**End Course Quiz Assessment**
	Group Learning LAB : Insight Canvas
	Capstone Project : DataStoryteller Challenge

Module 5: Applied Business Analytics

1. Descriptive Statistics	Learning Objective
	Probability Theory & Random Variables
	Summary Statistics
	Probability Distributions
	Skill Mastery Challenge
2. Inferential Statistics	Learning Objective
	Sampling & Statistical Inference
	Concept of Confidence
	Hypothesis testing
	Skill Mastery Challenge
Mid Course Assessment	**Mid Course Quiz Assessment**
	Group Learning LAB : StatMasters
3. Business Analysis	Learning Objective
	Business Metrics 101
	Web Analytics
	Experiement Design
	Workflow Management - Automated Code Execution
	Skill Mastery Challenge
Revision & Beyond	**Leveraging AI : Equations SImplified via ChatGPT**
	Research Paper - I
	Research paper - II
End Course Assessment	**End Course Quiz Assessment**
	Group Learning LAB : BizPulse
	Capstone Project : BizPulse Challenge

Module 6: Machine Learning & Generative AI with Microsoft Azure

1. Getting Started With ML	Learning Objective
	Mechanism behind Machine Learning
	Supervised learning - Regression
	Supervised learning - Classification
	Tree Based Model
	Bagging, Boosting & Non Linear Algorithms
	Data Preparation for ML Models & Cross validation
	Feature Engineering & Model Interpretation
	Skill Mastery Challenge
2. Advanced Algorithms	Learning Objective
	Unsupervised Learning
	Customer Segmentation - Case Study
	Time Series Analysis
	Recommender Systems
	Neural Networks
	Skill Mastery Challenge
Mid Course Assessment	**Mid Course Quiz Assessment**
	Group Learning LAB : AI Nexus
3. GenAI & MLOPS	Learning Objective
	ML LifeCycle & Overview of AI
	Data Pipeline, Flask & API
	ML Deployment with Microsoft Azure
	Computer Vision using Azure Studio
	Natual Language Processing
	Document Intelligence & Knowledge using Azure
	Generative AI and Prompt Engineering
	MultiPDFChat: Document Interaction Simplified
	Skill Mastery Challenge

Module 6: Machine Learning & Generative AI with Microsoft Azure contd.

Revision & Beyond	**Leveraging AI : Use AI to create AI**
	Research Paper - I
	Research paper - II
End Course Assessment	**End Course Quiz Assessment**
	Group Learning LAB : AI ModelOps
	Capstone Project : Intelligent Machines Challenge

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